The two–Helmholtz factorization and influence–matrix solve for the clamped wall–normal velocity equation

GPU DNS channel solver — design note

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Abstract

Per Runge–Kutta substep the wall–normal velocity $\hat{v}(y)$ of each Fourier mode (k_x, k_z) obeys a fourth–order, constant–coefficient two–point boundary value problem with four clamped conditions $\hat{v} = \mathrm{D}\hat{v} = 0$ at $y = \pm 1$. This note records, for later verification, the method we use to solve it on the GPU: factor the fourth–order operator into a product of two second–order Helmholtz operators, solve each with the banded Shen–Galerkin Helmholtz solver already validated, and recover the two missing boundary conditions with a 2×2 influence (capacitance) matrix per mode. The continuous algorithm, the boundary lifting used for the homogeneous solutions, the assembly of the influence matrix, the wall–derivative functionals, and the discrete sign conventions (matching `cheb_shen`) are all given explicitly. The dense direct solver `cheb_bih` remains the trusted reference against which this method is validated.

See also: `ZERO_TO_VETA.md` maps the end-to-end algorithm and the routine-level call graph — where this $\hat{v}$-solve sits in the per-step sequence and which routines carry it out.

1 The problem

After Crank–Nicolson treatment of the viscous term, the Kim–Moin–Moser $\hat{v}$–equation at stage k reduces, for each horizontal mode, to

$$\left(\mathcal{L} - \alpha\,\mathcal{L}^2\right)\hat{v} = \hat{f}, \qquad \mathcal{L} \equiv \mathrm{D}^2 - k^2, \qquad \alpha \equiv \frac{\beta_k\,\Delta t}{\mathrm{Re}} > 0, \tag{1}$$

with $\mathrm{D} \equiv \mathrm{d}/\mathrm{d}y$, $k^2 = k_x^2 + k_z^2 \geq 0$, on $y \in [-1, 1]$, subject to the four clamped boundary conditions

$$\hat{v}(-1) = \hat{v}(+1) = 0, \qquad \mathrm{D}\hat{v}(-1) = \mathrm{D}\hat{v}(+1) = 0. \tag{2}$$

The right–hand side $\hat{f}$ collects the explicit (nonlinear, previous–stage) forcing of the substep. Both $\mathcal{L}$ and $\hat{f}$ depend on the mode and the stage; α depends on the stage and on Δt but not on the mode.

2 Factorization into two Helmholtz operators

Because $\mathcal{L}$ is a constant–coefficient operator it commutes with itself and the quartic factors:

$$\mathcal{L} - \alpha\mathcal{L}^2 = \mathcal{L}\big(1 - \alpha\mathcal{L}\big) = -\alpha\,\mathcal{L}\Big(\mathcal{L} - \tfrac{1}{\alpha}\Big). \tag{3}$$

Define the two Helmholtz operators

$$\mathcal{H}_1 \equiv \mathcal{L} = \mathrm{D}^2 - \mu_1, \quad \mu_1 = k^2, \qquad \mathcal{H}_2 \equiv \mathcal{L} - \tfrac{1}{\alpha} = \mathrm{D}^2 - \mu_2, \quad \mu_2 = k^2 + \tfrac{1}{\alpha}. \tag{4}$$

Since $\alpha > 0$ we have $\mu_2 = k^2 + 1/\alpha > 0$ and $\mu_1 = k^2 \geq 0$; both operators are negative–definite under homogeneous Dirichlet data and hence boundedly invertible. Substituting (3) into (1),

$$\mathcal{H}_1 \mathcal{H}_2 \, \hat{v} = F, \qquad F \equiv -\frac{\hat{f}}{\alpha}. \tag{5}$$

$\mathcal{H}_1$ and $\mathcal{H}_2$ commute, so the order in which they are applied is immaterial; we take $\mathcal{H}_1$ as the outer operator.

3 The intermediate variable and its undetermined boundary data

Introduce the intermediate field

$$g \equiv \mathcal{H}_2 \, \hat{v} \quad \Longrightarrow \quad \mathcal{H}_1 \, g = F. \tag{6}$$

We have one second–order equation for g and one for $\hat{v}$, each of which admits exactly two boundary conditions. The two conditions $\hat{v}(\pm 1) = 0$ are imposed on the $\mathcal{H}_2$ solve. The $\mathcal{H}_1$ solve then requires boundary data for g, which is *not* given by the physics; we treat the two boundary values

$$\sigma_- \equiv g(-1), \qquad \sigma_+ \equiv g(+1) \tag{7}$$

as unknowns, to be fixed by the two remaining clamped conditions $\mathrm{D}\hat{v}(\pm 1) = 0$. This is the classical Kleiser–Schumann influence–matrix idea.

4 Linear superposition

By linearity, split both fields into a particular part (driven by F, with homogeneous boundary data) and two homogeneous parts (driven by unit boundary data, no interior forcing):

$$g = g_p + \sigma_- \, g_- + \sigma_+ \, g_+, \qquad \hat{v} = v_p + \sigma_- \, v_- + \sigma_+ \, v_+, \tag{8}$$

where the six fields solve

$$\mathcal{H}_1 g_p = F, \quad g_p(\pm 1) = 0, \tag{9}$$

$$\mathcal{H}_1 g_\pm = 0, \quad g_-(-1) = 1, \; g_-(+1) = 0, \quad g_+(-1) = 0, \; g_+(+1) = 1, \tag{10}$$

$$\mathcal{H}_2 v_p = g_p, \quad v_p(\pm 1) = 0, \tag{11}$$

$$\mathcal{H}_2 v_\pm = g_\pm, \quad v_\pm(\pm 1) = 0. \tag{12}$$

Each line is a homogeneous–Dirichlet Helmholtz solve except for the boundary data of $g_\pm$, handled by lifting in §5. With (9)–(12), g from (8) satisfies $\mathcal{H}_1 g = F$ and $g(\pm 1) = \sigma_\pm$, and $\hat{v}$ satisfies $\mathcal{H}_2 \hat{v} = g$; hence $\mathcal{H}_1 \mathcal{H}_2 \hat{v} = F$, i.e. (1), for *any* $\sigma_\pm$.

The Dirichlet conditions on $\hat{v}$ are automatic: since $v_p(\pm 1) = v_\pm(\pm 1) = 0$,

$$\hat{v}(\pm 1) = v_p(\pm 1) + \sigma_- v_-(\pm 1) + \sigma_+ v_+(\pm 1) = 0. \tag{13}$$

5 Boundary lifting for the homogeneous solutions

The fields $g_\pm$ carry inhomogeneous Dirichlet data and so cannot be represented directly in the homogeneous Shen–Dirichlet basis. Write

$$g_+ = \ell_+ + \tilde{g}_+, \qquad g_- = \ell_- + \tilde{g}_-, \tag{14}$$

with the linear lift functions carrying the boundary data,

$$\ell_+(y) = \tfrac{1}{2}(1+y) = \tfrac{1}{2}\big(T_0 + T_1\big), \qquad \ell_-(y) = \tfrac{1}{2}(1-y) = \tfrac{1}{2}\big(T_0 - T_1\big), \tag{15}$$

so that $\ell_+(+1) = 1,\ \ell_+(-1) = 0$ and $\ell_-(-1) = 1,\ \ell_-(+1) = 0$. Because each $\ell_\pm$ is linear, $\mathrm{D}^2\ell_\pm = 0$ and therefore $\mathcal{H}_1\ell_\pm = (\mathrm{D}^2 - k^2)\ell_\pm = -k^2\ell_\pm$. The corrections then solve homogeneous–Dirichlet Helmholtz problems

$$\mathcal{H}_1\,\tilde{g}_\pm = -\,\mathcal{H}_1\ell_\pm = k^2\,\ell_\pm, \qquad \tilde{g}_\pm(\pm 1) = 0, \tag{16}$$

which gives $\mathcal{H}_1 g_\pm = \mathcal{H}_1\ell_\pm + \mathcal{H}_1\tilde{g}_\pm = 0$ and the correct boundary data $g_\pm(\pm 1) = \ell_\pm(\pm 1)$ by construction.

6 The influence (capacitance) matrix

The only conditions left are $\mathrm{D}\hat{v}(\pm 1) = 0$. Differentiating (8) and evaluating at the walls,

$$\mathrm{D}\hat{v}(\pm 1) = \mathrm{D}v_p(\pm 1) + \sigma_-\,\mathrm{D}v_-(\pm 1) + \sigma_+\,\mathrm{D}v_+(\pm 1) \overset{!}{=} 0. \tag{17}$$

Collecting the two wall conditions gives the 2×2 system

$$\underbrace{\begin{pmatrix} \mathrm{D}v_-(-1) & \mathrm{D}v_+(-1) \\ \mathrm{D}v_-(+1) & \mathrm{D}v_+(+1) \end{pmatrix}}_{C} \begin{pmatrix} \sigma_- \\ \sigma_+ \end{pmatrix} = -\begin{pmatrix} \mathrm{D}v_p(-1) \\ \mathrm{D}v_p(+1) \end{pmatrix}. \tag{18}$$

The influence matrix C depends only on (k^2, α) through the homogeneous solutions $v_\pm$; it is assembled and inverted once per mode at setup. It is nonsingular whenever the clamped problem (1)–(2) is well posed, which holds for all $\alpha > 0$, $k^2 \geq 0$. With C^{-1} in hand,

$$\begin{pmatrix} \sigma_- \\ \sigma_+ \end{pmatrix} = -\,C^{-1}\begin{pmatrix} \mathrm{D}v_p(-1) \\ \mathrm{D}v_p(+1) \end{pmatrix}, \qquad C^{-1} = \frac{1}{\det C}\begin{pmatrix} C_{22} & -C_{12} \\ -C_{21} & C_{11} \end{pmatrix}. \tag{19}$$

7 Wall–derivative functionals

All wall conditions use the first derivative of a function given by its Chebyshev coefficients a_n, $u(y) = \sum_{n=0}^{N} a_n T_n(y)$. Using $T_n'(\pm 1) = (\pm 1)^{\,n-1}\,n^2$,

$$\mathrm{D}u(+1) = \sum_{n=0}^{N} n^2\,a_n, \qquad \mathrm{D}u(-1) = \sum_{n=0}^{N} (-1)^{\,n-1} n^2\,a_n. \tag{20}$$

These linear functionals are applied to the Chebyshev coefficients of v_p and $v_\pm$ (obtained from their Shen coefficients via `shen_to_cheb`).

8 Algorithm

Setup (once per mode, per stage; recompute when Δt changes).

1. Form the lift coefficients $\ell_\pm$ in Chebyshev space.
2. Solve (16) for $\tilde{g}_\pm$ (homogeneous Dirichlet, operator $\mathcal{H}_1$); set $g_\pm = \ell_\pm + \tilde{g}_\pm$ in Chebyshev space.
3. Solve (12) for $v_\pm$ (homogeneous Dirichlet, operator $\mathcal{H}_2$, RHS $g_\pm$); convert to Chebyshev space.
4. Evaluate $\mathrm{D}v_\pm(\pm 1)$ by (20); assemble C from (18) and store C^{-1}, together with $v_\pm$.

Per solve (per mode, per stage, per time step).

1. Solve $\mathcal{H}_1 g_p = F = -\hat{f}/\alpha$ with $g_p(\pm 1) = 0$; convert g_p to Chebyshev space.
2. Solve $\mathcal{H}_2 v_p = g_p$ with $v_p(\pm 1) = 0$; convert v_p to Chebyshev space.
3. Evaluate $\mathrm{D}v_p(\pm 1)$ by (20).
4. $\big(\sigma_-, \sigma_+\big)^\top = -\,C^{-1}\big(\mathrm{D}v_p(-1),\, \mathrm{D}v_p(+1)\big)^\top$.
5. $\hat{v} = v_p + \sigma_-\, v_- + \sigma_+\, v_+$.

Each solve costs two banded Helmholtz solves plus a handful of $O(n)$ reductions and a 2×2 apply; the two Helmholtz solves are exactly the validated banded GPU kernel, dispatched one mode per thread.

9 Discrete realization and sign conventions

The Shen–Dirichlet solver `helmholtz_solve`$(n, \mu, f_{\mathrm{cheb}})$ returns the Shen coefficients of the solution of

$$-u'' + \mu\, u = f, \qquad u(\pm 1) = 0, \tag{21}$$

i.e. of $(-\mathrm{D}^2 + \mu)u = f$. Our operators are $\mathcal{H}_i = \mathrm{D}^2 - \mu_i$, so $\mathcal{H}_i w = r$ is the same as $(-\mathrm{D}^2 + \mu_i)w = -r$. Hence the RHS sign flips on each call:

continuous solve	`helmholtz_solve` arguments	note
$\mathcal{H}_1 g_p = F = -\hat{f}/\alpha$	$\mu = k^2,\ \ f_{\mathrm{cheb}} = +\hat{f}/\alpha$	$-F = \hat{f}/\alpha$
$\mathcal{H}_2 v_p = g_p$	$\mu = k^2 + \frac{1}{\alpha},\ \ f_{\mathrm{cheb}} = -g_p$	
$\mathcal{H}_1 \tilde{g}_\pm = k^2 \ell_\pm$	$\mu = k^2,\ \ f_{\mathrm{cheb}} = -k^2 \ell_\pm$	
$\mathcal{H}_2 v_\pm = g_\pm$	$\mu = k^2 + \frac{1}{\alpha},\ \ f_{\mathrm{cheb}} = -g_\pm$	

The conversion $u_{\mathrm{shen}} \to a_{\mathrm{cheb}}$ uses $a_j = u_j - u_{j-2}$ (`shen_to_cheb`, the $\varphi_k = T_k - T_{k+2}$ basis). Both g_p and $g_\pm$ enter the subsequent $\mathcal{H}_2$ solve as Chebyshev–coefficient right–hand sides; the fact that $g_\pm$ is nonzero at the walls is irrelevant to the $\mathcal{H}_2$ solve, which constrains only the *solution* $v_\pm$.

10 Numerical remarks

Stiffness and cancellation. In the DNS regime $\alpha = \beta_k \Delta t/\mathrm{Re}$ is small, so $\mu_2 = k^2 + 1/\alpha$ is large and $\mathcal{H}_2$ is strongly mass–dominated (well conditioned). The intermediate $F = -\hat{f}/\alpha$ and g_p are large $O(1/\alpha)$, while $v_p = \mathcal{H}_2^{-1} g_p$ is $O(1)$: the $1/\alpha$ amplification cancels in the second solve. Because each individual Helmholtz solve is well conditioned, the factorization is numerically stable; the validation below confirms the recovered $\hat{v}$ matches the dense reference across α spanning the DNS range.

Variable time step. The mode–independent quantity is α (hence μ_2); the homogeneous solutions $v_\pm$ and the inverse influence matrix C^{-1} depend on (k^2, α) only. As with the Helmholtz operators, they are constant for fixed Δt and are recomputed whenever Δt (and therefore α, μ_2) changes – a setup–time operation, never inside the inner solve.

Validation. The method is checked mode by mode against the dense direct biharmonic solver `cheb_bih` (`biharmonic_solve`), comparing the recovered $\hat{v}$ in Chebyshev coefficients (via `shen_to_cheb` here and `shen_bih_to_cheb` there) across a sweep of n, k^2, and α. The GPU port then reuses the banded Helmholtz kernel for the two interior solves and a per–mode 2×2 apply for the boundary closure.